SmolWAM: A Latent World-Action Model for Robotic Manipulation on Edge Devices

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Abstract

World-action models (WAMs) predict a robot's actions together with the observations those actions will produce. However, the strongest models rely on video diffusion transformers that are far too computationally expensive for resource-constrained edge devices. We present SmolWAM, a latent WAM that we derive from the 2B-parameter Cosmos Policy by keeping its pretrained trunk and rebuilding its inference interface: single-frame visual latents, cached text features that preserve the trunk's attention sinks, 12 of 28 transformer blocks, and 1,008 tokens from which actions and future visual latents are read out in one rectified-flow step. Trained in 3.6 GPU-hours, SmolWAM succeeds in 95.5% of 400 LIBERO episodes at 57 ms, 2.3 TFLOP, and 2.4 GiB of GPU memory per 16-action chunk, 11× faster and 3.2× leaner than its teacher, and its world head still forecasts scene motion.

1 Introduction

World-action models (WAMs) predict a robot's next actions together with the observations they should produce [1–3]. Cosmos Policy [1] builds one from the Cosmos-Predict2 video model [4] by treating actions, future observations, and values as latent frames of a single denoising sequence, and reaches 98.5% average success on LIBERO [5]. Its inference stack, however, was built for video generation: a 4.8B-parameter T5-XXL text encoder [6], a VAE that encodes a 33-frame clip per query, and five denoising passes of a 2B-parameter transformer over 1,764 tokens, which cost 624 ms, 31 TFLOP, and 7.55 GiB per 16-action chunk on an A100 even with cached text features.

Efficient policies usually give up the world model or the video prior: compact vision-language-action (VLA) models such as SmolVLA [7] and TurboVLA [8] predict only actions, and Fast-WAM [9] skips future prediction at test time. We ask whether a strong pretrained WAM can itself be made small while still predicting the future.

We present SmolWAM (Fig. 1). Our premise is that much of the teacher's cost comes from an interface designed for video generation, not from what control requires. SmolWAM reuses the trunk's weights, token positions, and rectified-flow readout; it encodes single frames through a small adapter, replaces T5-XXL with cached BERT features [10] padded to preserve the trunk's attention sinks, keeps 12 of 28 blocks, and reads actions from 28 state-conditioned queries. One forward pass returns a 16-step action chunk and the next visual latents of both cameras. Most changes break the trunk zero-shot, so we distill them with a curriculum that recovers from pruning before compaction.

We contribute (i) SmolWAM, which retains 97% of its teacher's success at 11× lower latency; (ii) a method, with a diagnostic behind each step, for deriving compact WAMs from pretrained video WAMs (Sec. 3); and (iii) findings showing that the compressed world head preserves motion forecasting, whereas adding latent-action cascades or visual-reward fine-tuning degrades control (Sec. 4).

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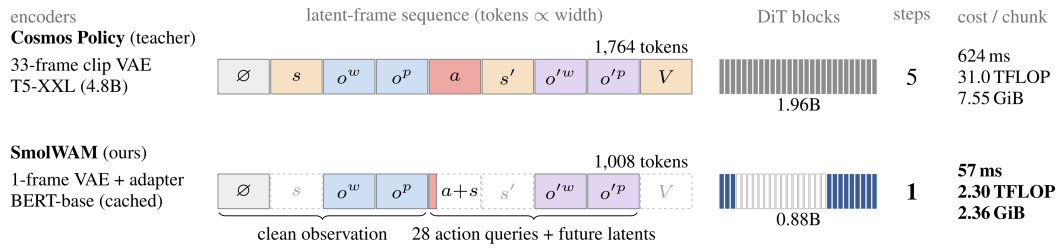


Figure 1: **From Cosmos Policy to SmolWAM.** Latent frames (width $\propto$ tokens), kept DiT blocks, denoising steps, and cost per 16-action chunk on an A100. SmolWAM drops the state and value frames (dashed), replaces the action frame with 28 state-conditioned queries ($a{+}s$), and reads actions and future camera latents (its world head) from one step.

2 Related Work

Efficient robot policies. Generalist VLAs bring pretraining to control [11–16], often with diffusion [17] or chunked [18] action heads; OpenVLA-OFT [19], TinyVLA [20], SmolVLA [7], and TurboVLA [8] speed them up; SmolWAM borrows TurboVLA's principles but retains a world model.

Predictive models for control. Latent world models learn dynamics in feature space [21–25], and GR-1 [26], PAD [27], and WorldVLA [28] generate future images and actions jointly. Recent WAMs build on video models [1, 2] and add action evaluation [29], modalities [30], object state [31], geometry [32], or latent actions [33]; Fast-WAM [9] drops future prediction at test time and LightWAM [34] decodes actions from a smaller video backbone, whereas SmolWAM reads future latents out of the pass that produces its actions.

3 SmolWAM

3.1 Teacher: Cosmos Policy

On LIBERO, the video diffusion transformer (DiT) [35] of Cosmos Policy (28 blocks, 1.96B parameters) denoises nine latent frames of $14{\times}14{=}196$ tokens (Fig. 1): a blank frame, the robot state s, the wrist and third-person camera latents o^w and o^p, a chunk a of $H{=}16$ actions, and, at time $t{+}H$, the state s', the camera latents o'^w and o'^p, and a value V. Images are encoded by the Wan2.1 VAE [36] from 33-frame clips, and the instruction enters through cross-attention to 512 T5-XXL embeddings. The model is a rectified flow [37, 38]: for $x_\tau = (1-\tau)\, x_0 + \tau\, \epsilon$ it predicts the velocity $v_\theta(x_\tau, \tau) \approx \epsilon - x_0$, which gives a closed-form clean estimate

$$\hat{x}_0 = x_\tau - \tau\, v_\theta(x_\tau, \tau). \tag{1}$$

Generated frames start at $\tau_{\max} \approx 0.988$ ($\sigma_{\max}{=}80$); the released pipeline takes five steps.

3.2 Deriving SmolWAM from the teacher

Single-frame latents. SmolWAM encodes each camera image once, as a single frame, with the Wan2.1 encoder (54M parameters). Such latents raise the teacher's action error (L1) from 0.0089 to 0.038; a 0.5M-parameter adapter maps them to clip-mode latents and restores it to 0.0092.

Cached text that keeps the attention sinks. We replace T5-XXL with cached BERT-base [10] features of the instruction, projected to the 1024-dimensional context. The trunk's cross-attention has bias-free key and value projections, so T5's $\sim$500 zero-padded positions contribute zero logits and values and act as an attention sink [39]: truncating the context to 32 or 128 positions drops the teacher to 0/5 closed-loop successes, and restoring the removed softmax mass analytically returns it to 5/5. SmolWAM thus zero-pads the BERT features to 512 positions; six zero-initialized vision-to-language layers (width 256), as in TurboVLA [8], let the instruction attend to the images.

Boundary-preserving pruning. Block influence $1 - \cos(h_{\text{in}}, h_{\text{out}})$ [40] is highest for blocks 0, 1, and 25–27, while blocks 3–22 are individually close to the identity. Zero-shot screens (5 episodes each) show that removals do not compose: skipping block 1, 26, or 27 fails every episode, skipping block 7, 14, or 20 alone fails none, and skipping four near-identity blocks (12–15) together fails all. SmolWAM keeps blocks 0–2 and 19–27 (0.88B parameters).

State-conditioned action queries. A zero-initialized MLP adds the robot state to 28 action queries that replace the 196-token action frame at its original positions. Beyond these queries we keep only the blank and image frames (1,008 tokens). Zero-shot, dropping the future-state and value frames barely changes the action error (0.009 to $\leq$0.019), but dropping the blank (0.30) or state (0.058) frame or compacting the vector frames to 28 tokens (0.21) does, so the layout is learned.

One-pass readout. SmolWAM evaluates its trunk once at $\tau_{\max}$ with fixed noise and reads the action chunk and future latents from Eq. (1); the teacher solves all 60 screening episodes in one step (Table 1). The future latents form SmolWAM's world head, supervised in training and never decoded at deployment.

3.3 Distillation with a layout curriculum

We train on 68,830 observations sampled every fourth step from the 1,700 successful LIBERO demonstrations released with Cosmos Policy, using the teacher's one-step predictions of the action chunk and future latents, a^T and z^T, cached at fixed noise (66 min on one A100). As in pruning and distillation of language models [41, 42], the loss $\|\hat{a}-a\|_1+\|\hat{a}-a^T\|_1+\|\hat{z}-z\|_1+\|\hat{z}-z^T\|_1$ matches the student's one-step estimates to both the demonstrations and the teacher, plus 0.1-weighted terms for the state and value frames while they exist. Pruning and compaction are separate distribution shifts, so the student, initialized from the kept teacher blocks, trains in the teacher's 1,764-token layout for the first 1,000 of 5,000 steps and in the compact layout afterward (AdamW; learning rate 2×10^{-5} for inherited and 10^{-4} for new weights; batch 32; bf16) in 3.6 hours on one A100.

4 Experiments

We ask (Q1) how much of the teacher's success SmolWAM retains, (Q2) where its savings come from, (Q3) whether it remains a world model, and (Q4) whether richer WAM components help.

Setup. LIBERO [5] has four suites of ten tasks; we use the teacher's observation pipeline (wrist and third-person images at 224 px, proprioception, 16-action chunks executed open loop), and 10 trials per task (400 episodes). A 60-episode screening set (3 tasks per suite, 5 trials each) includes the tasks an earlier pruned student failed most, so it scores below the full protocol. Latency is the wall-clock time from observation to action chunk on one A100-SXM4-40GB (batch 1, bf16), including preprocessing and VAE encoding, with text features cached for every model.

Q1: SmolWAM retains most of the teacher's success. SmolWAM succeeds in 382/400 episodes (95.5%; 95% Wilson interval 93.0–97.1) against the teacher's 393/400 (98.2%; 96.4–99.1). We detect no difference on Spatial and Object; the deficit lies in the multi-step Goal (94 vs. 99) and Long (89 vs. 97) suites, where failed rollouts usually complete the first subgoal and stall on the second. An earlier student that removed only 8 blocks and recovered with LoRA [43] distillation on 414 teacher observations reached 87.5%; SmolWAM differs in distillation coverage, full fine-tuning, and the learned layout, effects we have not separated. Training 7,500 more steps does not help (94.2%), and an 8-block variant falls to 83.3% on the screening set.

Q2: Where the 11$\times$ comes from. Running the unmodified teacher for one step on single-frame inputs cuts latency 5$\times$, to 126 ms, with no loss on the screening set; SmolWAM's trunk redesign adds 2.2$\times$ in latency, 3.3$\times$ in compute, and 1.9$\times$ in memory. At 57 ms per chunk (6.8 ms preprocessing, 12.9 ms VAE, 36.9 ms trunk) it issues 17.7 chunks (282 actions) per second; because chunks execute open loop, the chunk rate bounds how often the policy reacts to new observations. Per-channel INT8 quantization of the 172 largest linear layers shrinks the stored network from 1.64 to 0.85 GiB and costs one screening episode; simulated with bf16 kernels, it does not yet reduce latency.

Table 1: **LIBERO success and inference cost.** Screen: 60-episode screening set; suites and Full: 400 episodes. Cost per 16-action chunk on one A100 (batch 1, bf16); TFLOP counts trunk and VAE encoder, GiB is peak GPU memory. SmolWAM succeeds in 95.5% of episodes at 11× lower latency than its teacher. †Earlier evaluation pipeline (Cosmos Policy reports 98.5% [1]). ‡Shared GPU. §Simulated: INT8 weights, bf16 kernels.

	Inference cost				Success rate (%)					
Model	Trunk	TFLOP	ms	GiB	Screen	Spatial	Object	Goal	Long	Full
Cosmos Policy (5 steps)	1.96B	31.0	624	7.55	96.7	97.0	100.0	99.0	97.0	98.2†
Cosmos Policy (1 step, 1 frame)	1.96B	7.52	126	4.39	100.0	–	–	–	–	–
Pruned + LoRA distillation	1.40B	–	–	–	–	83.0	100.0	82.0	85.0	87.5†
SmolWAM	**0.88B**	**2.30**	**57**	**2.36**	93.3	99.0	100.0	94.0	89.0	**95.5**
+ 7.5k training steps	0.88B	2.30	60	2.36	93.3	94.0	100.0	94.0	89.0	94.2
INT8 weights§	0.88B	2.30	65‡	2.36	91.7	–	–	–	–	–
8 blocks	0.60B	1.76	134‡	1.84	83.3	–	–	–	–	–
SmolWAM-LA (4 steps)	0.92B	4.28	239	2.43	53.3	–	–	–	–	–

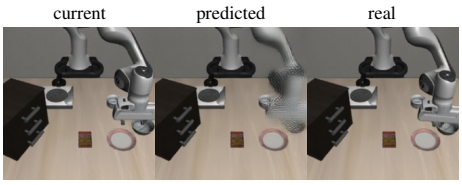


Figure 2: **World-head forecast**, decoded only for display.

Q3: SmolWAM remains a world model. On 96 held-out contexts, SmolWAM's forecast of the view 16 steps ahead lowers latent L1 by 24% relative to copying the current frame (0.101 vs. 0.133), and lowers the end-point error between RAFT [44] optical flow to the predicted and to the real future from 1.63 to 1.43 pixels. Decoded forecasts (Fig. 2) keep the scene and place the arm near its future pose but blur its appearance, and LPIPS [45] stays at the copy-frame level (0.128 vs. 0.115): the world head captures motion rather than texture.

Q4: Richer WAM components hurt at this scale. We port ABot-M0.5 [33] onto SmolWAM (39M more parameters): tokens from a frozen 32-dimensional latent-action model behind an asymmetric attention mask, mixture-of-transformer towers for the latent-action, arm, and gripper streams, four-step conditional flow matching, and dream forcing, which trains the action head on predicted latents. It succeeds in 53.3% of screening episodes at 239 ms (SmolWAM: 93.3% at 57 ms). Its one-step action estimate matches SmolWAM's offline (validation L1 0.089 vs. 0.091), but four-step sampling raises the error to 0.155 and one or two steps recover only 63.3% and 70.0%; its latent-action predictions are worse than predicting the mean code (MSE 1.15, variance 0.64), and its world head is no better (latent L1 0.102). Fine-tuning its world head with reward-weighted denoising regression under five rewards built from optical flow, depth [46], and LPIPS worsened every image metric (LPIPS from 0.130 to 0.167–0.177) and collapsed success to 5.0–16.7% within 300 steps, likely because the objective has no action term while the shared trunk trains.

5 Limitations and Conclusion

Limitations. We evaluate SmolWAM only on LIBERO; future work should extend evaluation to additional robotics benchmarks such as RoboCasa and, ultimately, real-robot settings. We also observe that reducing transformer depth disproportionately affects long-horizon tasks, with the largest degradation on LIBERO-Long. Developing pruning strategies that preserve long-horizon control performance therefore remains an important open problem.

Conclusion. A pretrained video WAM tolerates large interface and depth changes when each is distilled rather than applied zero-shot, but not auxiliary objectives that train its shared trunk without an action anchor; SmolWAM keeps its world prediction and 95.5% LIBERO success at a fraction of the teacher's cost.

References

[1] M. J. Kim, Y. Gao, T.-Y. Lin, Y.-C. Lin, Y. Ge, G. Lam, P. Liang, S. Song, M.-Y. Liu, C. Finn, and J. Gu. Cosmos Policy: Fine-Tuning Video Models for Visuomotor Control and Planning. In *International Conference on Learning Representations (ICLR)*, 2026.

[2] S. Ye, Y. Ge, K. Zheng, S. Gao, S. Yu, G. Kurian, S. Indupuru, Y. L. Tan, C. Zhu, J. Xiang, et al. World Action Models are Zero-shot Policies. *arXiv preprint arXiv:2602.15922*, 2026.

[3] Z. Lu, H. Zhai, G. Wang, H. Zeng, J. Yang, J. Liu, L. Cheng, Y. Zhang, Y. Qiu, Z. Cheng, I. Laptev, D. Xu, B. Riviere, G. Loianno, E. Xing, and X. Zuo. World-Action Models for Robot Learning and Control: A Survey. *arXiv preprint arXiv:2609.16074*, 2026.

[4] NVIDIA, N. Agarwal, A. Ali, M. Bala, Y. Balaji, E. Barker, T. Cai, P. Chattopadhyay, Y. Chen, Y. Cui, et al. Cosmos World Foundation Model Platform for Physical AI. *arXiv preprint arXiv:2501.03575*, 2025.

[5] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone. LIBERO: Benchmarking Knowledge Transfer for Lifelong Robot Learning. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, volume 36, pages 44776–44791, 2023.

[6] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.

[7] M. Shukor, D. Aubakirova, F. Capuano, P. Kooijmans, S. Palma, A. Zouitine, M. Aractingi, C. Pascal, M. Russi, A. Marafioti, S. Alibert, M. Cord, T. Wolf, and R. Cadene. SmolVLA: A Vision-Language-Action Model for Affordable and Efficient Robotics. *arXiv preprint arXiv:2506.01844*, 2025.

[8] H. Xie, C. Yao, X. Wu, Y. Zhu, D. Liang, X. Bai, and H. Ding. TurboVLA: Real-Time Vision-Language-Action Model at 32 Hz on an RTX 4090 with $<$1 GB VRAM. *arXiv preprint arXiv:2607.27205*, 2026.

[9] T. Yuan, Z. Dong, Y. Liu, and H. Zhao. Fast-WAM: Do World Action Models Need Test-time Future Imagination? *arXiv preprint arXiv:2603.16666*, 2026.

[10] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT), Volume 1 (Long and Short Papers)*, pages 4171–4186, 2019.

[11] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, X. Chen, K. Choromanski, T. Ding, D. Driess, A. Dubey, C. Finn, et al. RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control. *arXiv preprint arXiv:2307.15818*, 2023.

[12] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. P. Foster, P. R. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, S. Levine, P. Liang, and C. Finn. OpenVLA: An Open-Source Vision-Language-Action Model. In *Proceedings of the 8th Conference on Robot Learning (CoRL)*, volume 270 of *Proceedings of Machine Learning Research*, pages 2679–2713, 2024.

[13] Open X-Embodiment Collaboration, A. O'Neill, A. Rehman, A. Maddukuri, A. Gupta, A. Padalkar, A. Lee, A. Pooley, A. Gupta, A. Mandlekar, et al. Open X-Embodiment: Robotic Learning Datasets and RT-X Models. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 6892–6903, 2024.

[14] D. Ghosh, H. R. Walke, K. Pertsch, K. Black, O. Mees, S. Dasari, J. Hejna, T. Kreiman, C. Xu, J. Luo, Y. L. Tan, L. Y. Chen, Q. Vuong, T. Xiao, P. R. Sanketi, D. Sadigh, C. Finn, and S. Levine. Octo: An Open-Source Generalist Robot Policy. In *Proceedings of Robotics: Science and Systems (RSS)*, 2024.

[15] K. Black, N. Brown, D. Driess, A. Esmail, M. R. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter, S. Jakubczak, T. Jones, L. Ke, S. Levine, A. Li-Bell, M. Mothukuri, S. Nair, K. Pertsch, L. X. Shi, L. Smith, J. Tanner, Q. Vuong, A. Walling, H. Wang, and U. Zhilinsky. π_0: A Vision-Language-Action Flow Model for General Robot Control. In *Proceedings of Robotics: Science and Systems (RSS)*, 2025.

[16] K. Pertsch, K. Stachowicz, B. Ichter, D. Driess, S. Nair, Q. Vuong, O. Mees, C. Finn, and S. Levine. FAST: Efficient Action Tokenization for Vision-Language-Action Models. In *Proceedings of Robotics: Science and Systems (RSS)*, 2025.

[17] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. C. Burchfiel, and S. Song. Diffusion Policy: Visuomotor Policy Learning via Action Diffusion. In *Proceedings of Robotics: Science and Systems (RSS)*, 2023.

[18] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn. Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware. In *Proceedings of Robotics: Science and Systems (RSS)*, 2023.

[19] M. J. Kim, C. Finn, and P. Liang. Fine-Tuning Vision-Language-Action Models: Optimizing Speed and Success. In *Proceedings of Robotics: Science and Systems (RSS)*, 2025.

[20] J. Wen, Y. Zhu, J. Li, M. Zhu, Z. Tang, K. Wu, Z. Xu, N. Liu, R. Cheng, C. Shen, Y. Peng, F. Feng, and J. Tang. TinyVLA: Toward Fast, Data-Efficient Vision-Language-Action Models for Robotic Manipulation. *IEEE Robotics and Automation Letters*, 10(4):3988–3995, 2025.

[21] G. Zhou, H. Pan, Y. LeCun, and L. Pinto. DINO-WM: World Models on Pre-trained Visual Features enable Zero-shot Planning. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pages 79115–79135, 2025.

[22] M. Assran, A. Bardes, D. Fan, Q. Garrido, R. Howes, M. Komeili, M. Muckley, A. Rizvi, C. Roberts, K. Sinha, et al. V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. *arXiv preprint arXiv:2506.09985*, 2025.

[23] R. Zheng, J. Wang, S. Reed, J. Bjorck, Y. Fang, F. Hu, J. Jang, K. Kundalia, Z. Lin, L. Magne, A. Narayan, Y. L. Tan, G. Wang, Q. Wang, J. Xiang, Y. Xu, S. Ye, J. Kautz, F. Huang, Y. Zhu, and L. Fan. FLARE: Robot Learning with Implicit World Modeling. In *Proceedings of the 9th Conference on Robot Learning (CoRL)*, volume 305 of *Proceedings of Machine Learning Research*, pages 3952–3971, 2025.

[24] L. Maes, Q. Le Lidec, D. Scieur, Y. LeCun, and R. Balestriero. LeWorldModel: Stable End-to-End Joint-Embedding Predictive Architecture from Pixels. *arXiv preprint arXiv:2603.19312*, 2026.

[25] Y. Gao and X. Xu. Fast LeWorldModel. *arXiv preprint arXiv:2606.26217*, 2026.

[26] H. Wu, Y. Jing, C. Cheang, G. Chen, J. Xu, X. Li, M. Liu, H. Li, and T. Kong. Unleashing Large-Scale Video Generative Pre-training for Visual Robot Manipulation. In *International Conference on Learning Representations (ICLR)*, 2024.

[27] Y. Guo, Y. Hu, J. Zhang, Y.-J. Wang, X. Chen, C. Lu, and J. Chen. Prediction with Action: Visual Policy Learning via Joint Denoising Process. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, pages 112386–112410, 2024.

[28] J. Cen, C. Yu, H. Yuan, Y. Jiang, S. Huang, J. Guo, X. Li, Y. Song, H. Luo, F. Wang, D. Zhao, and H. Chen. WorldVLA: Towards Autoregressive Action World Model. *arXiv preprint arXiv:2506.21539*, 2025.

[29] P. Zhou, S. Chen, D. Chen, J. Wang, R. Jin, B. Zhu, Y. Pan, S. Gu, K. Wang, S. Nan, X. Qiu, C. Qiu, P. Yang, Y. Cai, J. Gao, Y. Li, Y. Fu, X. Yue, Z. Chen, and J. Luo. τ_0-WM: A Unified Video-Action World Model for Robotic Manipulation. *arXiv preprint arXiv:2606.01027*, 2026.

[30] NVIDIA, Aditi, N. Agarwal, A. Ali, J. Allen, M. Antolini, A. Aubame, A. Azzolini, J. Bai, M. Bala, et al. Cosmos 3: Omnimodal World Models for Physical AI. *arXiv preprint arXiv:2606.02800*, 2026.

[31] Y. Liu, P. Sun, S. Li, Y. Xie, L. Zhang, X. Chao, S. Dong, F. Chen, X.-P. Zhang, and W. Ding. OA-WAM: Object-Addressable World Action Model for Robust Robot Manipulation. *arXiv preprint arXiv:2605.06481*, 2026.

[32] J. A. Lopetegui Gonzalez, P. Pacaud, and C. Schmid. Spatially Aware World Action Model via Geometric Latent Diffusion. *arXiv preprint arXiv:2609.02531*, 2026.

[33] R. Chen, Y. Yang, Z. Tang, D. Huo, T. Lin, H. Wu, H. Liu, Y. Chen, L. Zheng, B. Yuan, T. Li, M. Wang, D. Qi, B. Hu, W. Mei, Y. Xuan, H. Yang, Y. Zhu, M. Xu, Z. Ma, and X. Chang. ABot-M0.5: Unified Mobility-and-Manipulation World Action Model. *arXiv preprint arXiv:2607.00678*, 2026.

[34] Z. Li, D. Cheng, Y. Wang, S. Wang, X. Xu, L. Weng, J. Wang, and J. Wang. Light-WAM: Efficient World Action Models with State-Fusion Action Decoding. *arXiv preprint arXiv:2606.08242*, 2026.

[35] W. Peebles and S. Xie. Scalable Diffusion Models with Transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 4195–4205, 2023.

[36] Team Wan, A. Wang, B. Ai, B. Wen, C. Mao, C.-W. Xie, D. Chen, F. Yu, H. Zhao, J. Yang, et al. Wan: Open and Advanced Large-Scale Video Generative Models. *arXiv preprint arXiv:2503.20314*, 2025.

[37] X. Liu, C. Gong, and Q. Liu. Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow. In *International Conference on Learning Representations (ICLR)*, 2023.

[38] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. Flow Matching for Generative Modeling. In *International Conference on Learning Representations (ICLR)*, 2023.

[39] G. Xiao, Y. Tian, B. Chen, S. Han, and M. Lewis. Efficient Streaming Language Models with Attention Sinks. In *International Conference on Learning Representations (ICLR)*, 2024.

[40] X. Men, M. Xu, Q. Zhang, Q. Yuan, B. Wang, H. Lin, Y. Lu, X. Han, and W. Chen. ShortGPT: Layers in Large Language Models are More Redundant Than You Expect. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 20192–20204, 2025.

[41] S. Muralidharan, S. Turuvekere Sreenivas, R. Joshi, M. Chochowski, M. Patwary, M. Shoeybi, B. Catanzaro, J. Kautz, and P. Molchanov. Compact Language Models via Pruning and Knowledge Distillation. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, pages 41076–41102, 2024.

[42] G. Hinton, O. Vinyals, and J. Dean. Distilling the Knowledge in a Neural Network. *arXiv preprint arXiv:1503.02531*, 2015.

[43] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. LoRA: Low-Rank Adaptation of Large Language Models. In *International Conference on Learning Representations (ICLR)*, 2022.

[44] Z. Teed and J. Deng. RAFT: Recurrent All-Pairs Field Transforms for Optical Flow. In *European Conference on Computer Vision (ECCV)*, pages 402–419. Springer, 2020.

[45] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang. The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

[46] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, and H. Zhao. Depth Anything V2. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, pages 21875–21911, 2024.